DYNAMIC DISTILLATION / COMPLETED RUN

C3C4_QR_xB_C3_control_continue2700s_QR65_20260716

Continue_bottoms_C3_control_45min_with_QR_limit_65MMBtuhr

RUN VALIDITY	DYNAMIC GATE	FINAL SCORE	SIM / WALL
REVIEW	**PASS**	**0.913**	**0.180**

AUTOMATED ASSESSMENT Top pressure is 2.32 psi from its logged target, so the rate gate does not establish acceptable pressure control.

Run Identity

Item	Value
Run ID	20260716_095832
Started	2026-07-16 09:58:32
Completed	2026-07-16 14:07:56
Run type	Continuation from native checkpoint
Input workbook	C:\Users\Thomas Zvolensky\Documents\Python Scripts\Dynamic_DistillationII\logs\c3c4_initializer_residual_vapor_state_stage2_20260706.xlsx
Input checkpoint	C:\Users\Thomas Zvolensky\Documents\Python Scripts\Dynamic_DistillationII\logs\c3c4_QR_xB_C3_control_continue300s_20260716\c3c4_initializer_residual_vapor_state_stage2_20260706__checkpoint_20260716_092914.npz
Summary data	C:\Users\Thomas Zvolensky\Documents\Python Scripts\Dynamic_DistillationII\logs\c3c4_QR_xB_C3_control_continue2700s_QR65_20260716\column_summary_20260716_095832.csv
Tray-profile data	C:\Users\Thomas Zvolensky\Documents\Python Scripts\Dynamic_DistillationII\logs\c3c4_QR_xB_C3_control_continue2700s_QR65_20260716\column_profile_20260716_095832.csv

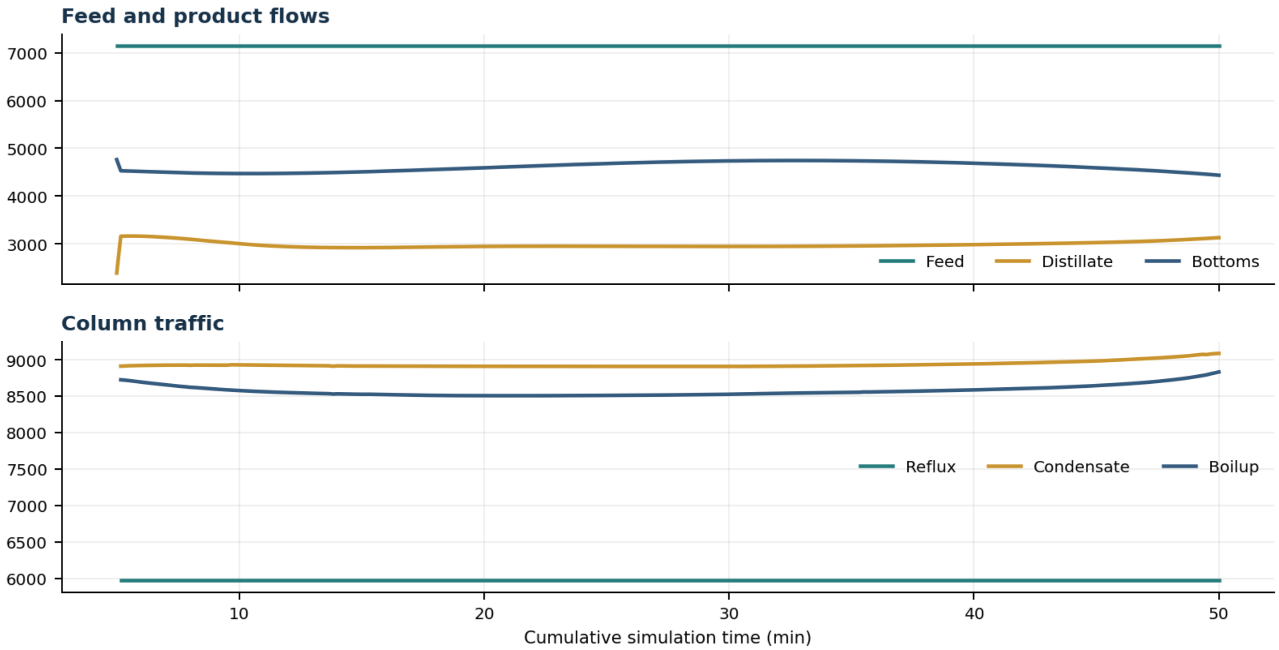
Operating Snapshot

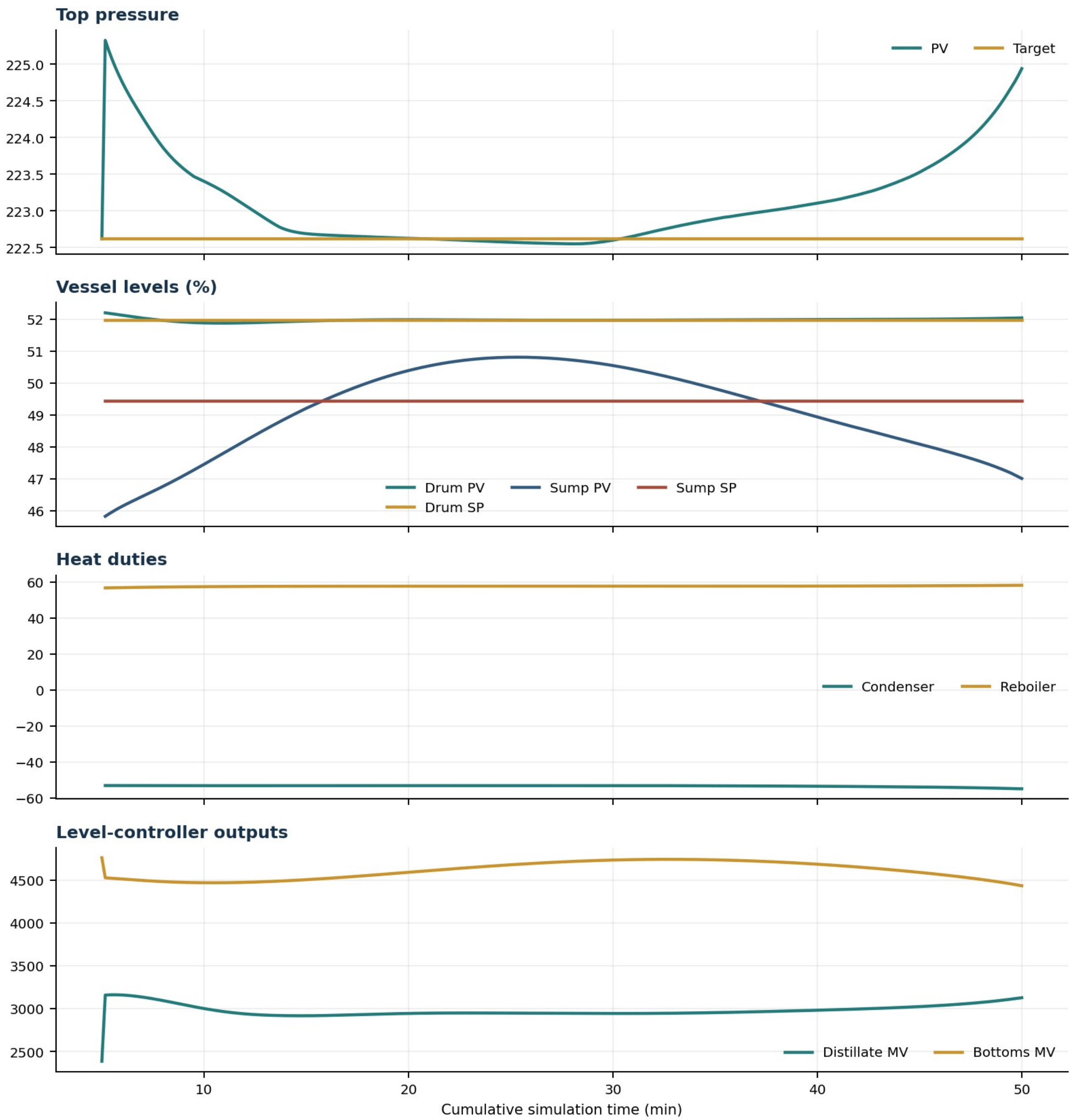
Parameter	Start	End	Units
Feed	7,142.98	7,142.98	lbmol/h
Distillate	2,386.93	3,128.20	lbmol/h
Bottoms	4,761.98	4,434.96	lbmol/h
Reflux	Not reported	5,967.32	lbmol/h
Top pressure	222.619	224.939	psia
Bottom pressure	314.167	232.263	psia
Distillate temperature	113.43	113.43	deg F
Bottoms temperature	208.01	213.65	deg F
Condenser duty	Not reported	-55.000	MMBtu/h
Reboiler duty	Not reported	58.250	MMBtu/h
Distillate drum controller PV	Not reported	52.04	lbmol
Bottoms sump controller PV	Not reported	47.01	lbmol
Steady-state score	Not reported	0.9127	-

Product Composition

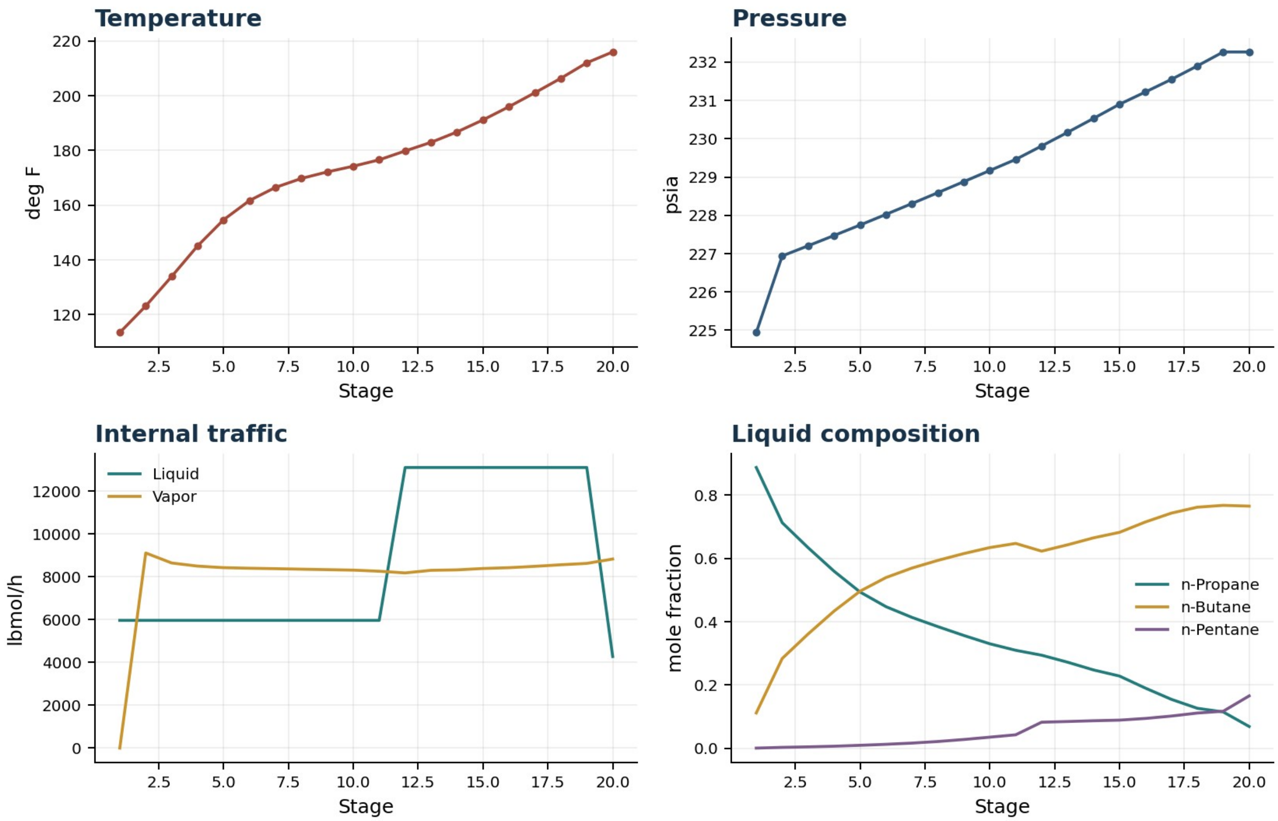
Component	Distillate start	Distillate end	Bottoms start	Bottoms end
n-Butane	6.379%	8.427%	72.356%	77.765%
n-Pentane	0.005%	0.015%	15.796%	16.713%
n-Propane	93.616%	91.559%	11.848%	5.522%
source	Not reported	Not reported	Not reported	Not reported

Dynamic Trends




Top pressure
225.0
224.5
224.0
223.5
223.0
222.5
PV
Target
Vessel levels (%)
52
51
50
49
48
47
46
Drum PV
Sump PV
Sump SP
Drum SP
Heat duties
60
40
20
0
−20
−40
−60
Condenser
Reboiler
Level-controller outputs
4500
4000
3500
3000
2500
Distillate MV
Bottoms MV
10
20
30
40
50
Cumulative simulation time (min)

Final Tray Profiles



Final Tray Profile Table

Stage	T (deg F)	P (psia)	L (lbmol/h)	V (lbmol/h)	x n-Propane	x n-Butane	x n-Pentane	y n-Propane	y n-Butane	y n-Pentane
1	113.43	224.939	5,967.3	0.0	0.8877	0.1117	0.0006	0.0000	0.0000	0.0000
2	123.14	226.939	5,967.3	9,116.4	0.7131	0.2839	0.0030	0.8877	0.1117	0.0006
3	133.94	227.204	5,967.3	8,647.7	0.6340	0.3615	0.0045	0.7375	0.2592	0.0033
4	145.12	227.473	5,967.3	8,502.2	0.5597	0.4337	0.0066	0.6754	0.3194	0.0052
5	154.66	227.746	5,967.3	8,429.7	0.4941	0.4966	0.0094	0.6183	0.3739	0.0078
6	161.69	228.024	5,967.3	8,402.6	0.4477	0.5397	0.0126	0.5695	0.4195	0.0110
7	166.49	228.306	5,967.3	8,383.0	0.4138	0.5697	0.0165	0.5347	0.4507	0.0146
8	169.76	228.591	5,967.3	8,358.7	0.3846	0.5939	0.0215	0.5089	0.4724	0.0187
9	172.16	228.878	5,967.3	8,337.4	0.3567	0.6155	0.0278	0.4862	0.4903	0.0234
10	174.24	229.167	5,967.3	8,314.4	0.3304	0.6344	0.0352	0.4643	0.5068	0.0289
11	176.57	229.458	5,967.3	8,263.5	0.3098	0.6476	0.0427	0.4447	0.5209	0.0344
12	179.82	229.809	13,110.3	8,184.8	0.2943	0.6232	0.0826	0.4311	0.5301	0.0388
13	182.97	230.165	13,110.3	8,306.2	0.2721	0.6432	0.0847	0.3860	0.5714	0.0427
14	186.77	230.528	13,110.3	8,327.0	0.2477	0.6652	0.0871	0.3525	0.6017	0.0457
15	191.16	230.897	13,110.3	8,391.0	0.2284	0.6826	0.0890	0.3165	0.6341	0.0494
16	195.99	231.218	13,110.3	8,428.3	0.1903	0.7154	0.0944	0.2910	0.6569	0.0521
17	201.09	231.546	13,110.3	8,491.9	0.1548	0.7433	0.1019	0.2366	0.7031	0.0602
18	206.36	231.898	13,110.3	8,564.8	0.1265	0.7621	0.1114	0.1865	0.7416	0.0719
19	212.04	232.263	13,110.3	8,629.0	0.1149	0.7679	0.1172	0.1486	0.7652	0.0862
20	216.00	232.263	4,277.1	8,833.2	0.0689	0.7655	0.1656	0.1372	0.7691	0.0937

Simulation Configuration

Parameter	Value
Runtime Mode	hydraulic
Thermo Mode	dwsim
Dwsim Property Package	pr
Integrator	explicit-euler
N Steps	13500
Dt Sec	0.2
Log Every N Steps	50
Thermo Every N Steps	1
Include Energy	True
Enable Equilibrium Relaxation	True
Equilibrium Relaxation Mode	composition-exponential
Equilibrium Tau Sec	0.5
Flash Feed At Stage Conditions	True
Enable Liquid Hydraulic Override	True
Liquid Hydraulic Model	francis
Liquid Hydraulic Override Alpha	0.25
Enable Level Control	True
Top Level Pv Mode	true-level
Top Level Kc	20.0
Top Level Ti Sec	120.0
Bottom Level Pv Mode	true-level
Bottom Level Kc	3.0
Bottom Level Ti Sec	300.0
Enable Pressure Control	True
Pressure Control Mv	condenser-duty
Top Pressure Sp Psia	222.6194497041178
Top Pressure Kc	-300000.0
Top Pressure Ti Sec	180.0
Top Pressure Pv Filter Tau Sec	5.0
Top Pressure Mv Slew Limit Per S	25000.0
Condenser Duty Mode	specified
Condenser Duty Btu Per H	-51600000.0
Condenser Duty Min Btu Per H	-55000000.0
Condenser Duty Max Btu Per H	-46000000.0
Reboiler Duty Btu Per H	49340000.0
Vapor Holdup Relaxation Sec	0.0
Vapor Flow Relaxation Sec	0.0
Vapor Flow Zero Temperature Target	True
Dynamic Vflow Nominal Hi Ratio	1.05
Steady State Window Sec	30.0
Steady State Min Time Sec	60.0
Steady State Rel State Rate Tol Per S	0.003
Steady State Temp Rate Tol F Per S	0.15
Steady State Sp Error Tol	0.02
Steady State Require Sp	False

Exact Launch Command

"C:\Users\Thomas Zvolensky\anaconda3\python.exe" -m dynamic_distillation.dynamic_run_scaffold_v1 --excel logs\c3c4_initializer_residual_vapor_state_stage2_20260706.xlsx --init-from-checkpoint "C:\Users\Thomas Zvolensky\Documents\Python Scripts\Dynamic_DistillationII\logs\c3c4_QR_xB_C3_control_continue300s_20260716\c3c4_initializer_residual_vapor_state_stage2_20260706__checkpoint_20260716_092914.npz" --run-name C3C4_QR_xB_C3_control_continue2700s_QR65_20260716 --run-description Continue_bottoms_C3_control_45min_with_QR_limit_65MMBtuhr --runtime-mode hydraulic --thermo dwsim --dwsim-property-package pr --include-energy --equilibrium-relaxation-mode composition-exponential --equilibrium-tau-sec 0.5 --flash-feed-at-stage-conditions --enable-liquid-hydraulic-override --liquid-hydraulic-model francis --liquid-hydraulic-override-alpha 0.25 --enable-level-control --ignore-workbook-level-pv-mode --top-level-pv-mode true-level --bottom-level-pv-mode true-level --top-level-kc 20 --top-level-ti 120 --bottom-level-kc 3 --bottom-level-ti 300 --enable-pressure-control --pressure-control-mv condenser-duty --top-pressure-sp 222.6194497041178 --top-pressure-kc -300000 --top-pressure-ti 180 --top-pressure-pv-filter-tau-sec 5 --top-pressure-mv-slew-limit-per-s 25000 --condenser-duty-mode specified --condenser-duty-btuph -51600000 --condenser-duty-min-btuph -55000000 --condenser-duty-max-btuph -46000000 --enable-top-drum-resident-condensation --top-drum-resident-condensation-tau-sec 30 --top-drum-resident-condensation-max-frac-per-step 0.10 --reflux 5967.322711692588 --vapor-holdup-relaxation-sec 0 --vapor-flow-relaxation-sec 0 --vapor-flow-zero-temperature-target --use-excel-vapor-holdup --dynamic-vflow-nominal-hi-ratio 1.05 --disable-startup-thermo-conditioning --disable-restart-reentry-settling --enable-bottoms-composition-control --bottoms-comp-component C3 --bottoms-comp-sp 0.04717 --bottoms-comp-kc 50000000 --bottoms-comp-ti 600 --bottoms-comp-mv reboiler-duty --reboiler-duty-btuph 49340000 --reboiler-duty-cmd-min-btuph 45000000 --reboiler-duty-cmd-max-btuph 65000000 --n-steps 13500 --dt 0.2 --log-every 50 --logs-dir logs\c3c4_QR_xB_C3_control_continue2700s_QR65_20260716 --no-word-report --allow-repeat-command